



SENIOR AI ENGINEER CHALLENGE

Distribution: Candidates

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GOAL

For the Senior AI Engineer challenge, we are providing you with 1 use case. We would like to learn how you would solve this with AI. We want you to come up with a basic solution and implement it. You are presenting towards the 'lead engineers of the Customer'.

What will the interview look like?

[keep an eye on time management!]

- [5min] Quick round table for everyone to introduce themselves
- [20min] Presentation on your design and implementation
You can choose if you'd like to create a presentation for it or use another format (website, whiteboard, ...), we would like to see/hear how you approached it in a digestible format. If you decide to present slides, make sure to keep it limited to 10 slides.
- [25min] Q&A
- [10min] Questions from your side

Case - Automated Generative Marketing Collateral

Customer C is a marketing automation agency that creates personalized B2B marketing materials (like physical newsletters and brochures). Currently, this process is highly manual, tedious, and difficult to scale. When a 'Sender' (e.g., an AI software vendor) wants to target a 'Receiver' (e.g., a large logistics company), human editors perform the following steps manually:

- **Research:** Visit both companies' websites to understand the Sender's offerings and the Receiver's industry challenges.
- **Content Creation:** Write custom articles from scratch that bridge these two domains, ensuring the tone is appropriate and the narrative makes sense for a B2B pitch.

ML6

- **Asset Gathering:** Manually source relevant visuals, such as extracting company logos or finding contextually appropriate images.
- **Layout Formatting:** Meticulously copy-paste all this gathered text and imagery into strict, pre-defined publishing templates. This often involves trial and error to ensure the text fits perfectly within column limits, font sizes are adhered to, and no awkward white space is left behind.

They want to fully automate this pipeline using Generative AI. The desired system should take as input context PDFs on both the sender and receiver company and use an LLM to generate tailored, factually correct articles. The input PDFs contain text, images and tables. Finally, the system must output a JSON that maps the generated text into a pre-defined layout template while respecting specific constraints (e.g., word limits, image placeholders, and theme colors).

YOUR TASK:

Design a technical solution to this challenge. Your design must contain the orchestration of the contextual data retrieval, LLM interaction and layout formatting. It must also contain a cloud architecture for either AWS, GCP or Azure.

Implement:

Build a lightweight prototype backend. Your prototype must expose two endpoints:

- An endpoint to upload PDFs as context for a certain sender and receiver company.
- An endpoint to generate a short, tailored article as a structured JSON based on a prompt for a certain sender and receiver company.

Deliverable:

- Code Submission: please upload your code to a **private** GitHub repository. After the challenge we might ask you to share your code with your interviewers. You don't need to share the code before the interview.
- You will walk the lead engineers of the Customer through your solution.